

Rearranging and removing repeated terms, we get:

$$F = xy'z + x'y'z + x'yz' + xy'z + xy'z$$

$$\Sigma(0,1,4,5)$$

$$\begin{aligned} d &= yz + xy = yz \cdot 1 + xy \cdot 1 \text{ (Postulate 2(b))} \\ &= yz \cdot (x + x') + xy \cdot (z + z') \text{ (Postulate 5(a))} \\ &= yzx + yzx' + xyz + xyz' \text{ (Postulate 4(a))} \end{aligned}$$

Rearranging and removing repeated terms, we get:

$$d = x'yz + xyz' + xyz \quad \Sigma(3,6,7)$$



Simplified SOP: 1

$$\begin{aligned} b) F &= B'C'D' + BCD' + ABCD' \\ d &= B'C'D' + ABC'D' \end{aligned}$$

Solution:

$$\begin{aligned} F &= B'C'D' \cdot 1 + BCD' \cdot 1 + ABCD' \text{ (Postulate 2(b))} \\ &= B'C'D' \cdot (A + A') + BCD' \cdot (A + A') + ABCD' \text{ (Postulate 5(a))} \\ &= B'C'D'A + B'C'D'A' + BCD'A + BCD'A' + ABCD' \text{ (Postulate 4(a))} \end{aligned}$$

Rearranging and removing repeated terms.

$$F = A'B'C'D' + A'BCD' + AB'C'D' + ABCD'$$

$$\Sigma(0,6,8,14)$$

$$\begin{aligned} d &= B'C'D' \cdot 1 + A'BC'D' \text{ (Postulate 2(b))} \\ &= B'C'D' \cdot (A + A') + A'BC'D' \text{ (Postulate 5(a))} \\ &= B'C'D'A + B'C'D'A' + A'BC'D' \text{ (Postulate 4(a))} \end{aligned}$$

Rearranging and removing repeated terms

$$d = A'BC'D' + A'BC'D' + A'BC'D'$$

$$\Sigma(2,5,10)$$